$$\prod_{1\le i<j\le k} R^{(ij)}(i-j) = \prod_{1\le i<j\le k}\left(1-\frac{P_{ij}}{j-i}\right)$$

$$(*)\quad a_k = \sum_{\sigma\in S_k}(-1)^{\sigma}\sigma = \prod_{1\le i\le k-1}\ \prod_{i+1\le j\le k}\left(1-\frac{P_{ij}}{j-i}\right)$$

b_k

$k=2$

$$b_2 = 1-\frac{P_{12}}{2-1} = 1-P_{12} = a_2$$

Let $a_{k-1} = b_{k-1}$

Then $a_k = (1-P_{1k}-\cdots-P_{k-1,k})\,a_{k-1}$

$$b_k = \left[1-\frac{P_{k-1,k}}{k-(k-1)}\right]\left[\left(1-\frac{P_{k-2,k}}{k-(k-2)}\right)\left(1-\frac{P_{k-1,k}}{k-(k-1)}\right)\right]\cdots\left[\left(1-\frac{P_{1k}}{k-1}\right)\left(1-\frac{P_{1,k-1}}{k-2}\right)\cdots\left(1-\frac{P_{12}}{1}\right)\right]$$

$$= \left[\prod_{1\le i\le k-1}\left(1-\frac{P_{ik}}{k-i}\right)\right]b_{k-1} = \left[\prod_{1\le i\le k-1}\left(1-\frac{P_{ik}}{k-i}\right)\right]a_{k-1}$$

$$= \prod_{2\le i\le k-1}\left(1-\frac{P_{ik}}{k-i}\right)\left(1-\frac{P_{1k}}{k-1}\right)$$

$$= \prod_{2\le i\le k-1}\left(1-\frac{P_{ik}}{k-i}\right)a_{k-1} - \frac{1}{k-1}\prod_{2\le i\le k-1}\left(1-\frac{P_{ik}}{k-i}\right)P_{1k}\,a_{k-1}$$

For $i = 2,\ldots,k-1$, $P_{ik}P_{1k} = P_{1k}P_{i1}$

$$\Rightarrow \frac{1}{k-1}\prod_{2\le i\le k-1}\left(1-\frac{P_{ik}}{k-i}\right)P_{1k}\,a_{k-1} = \frac{P_{1k}}{k-1}\left(1-\frac{P_{k-1,1}}{1}\right)\left(1-\frac{P_{k-2,1}}{2}\right)\cdots\left(1-\frac{P_{21}}{k-2}\right)a_{k-1}$$

$$= \frac{P_{1k}}{k-1}\left(1+\frac{1}{1}\right)\left(1+\frac{1}{2}\right)\cdots\left(1+\frac{1}{k-2}\right)a_{k-1}$$

$$= P_{1k}\,a_{k-1}$$

$(P_{i1}\,a_{k-1} = -a_{k-1})$

$$\Rightarrow b_k = \prod_{2\le i\le k-1}\left(1-\frac{P_{ik}}{k-i}\right)a_{k-1} - P_{1k}\,a_{k-1} = \cdots = \left(1-P_{k-1,k}-P_{k-2,k}-\cdots-P_{1k}\right)a_{k-1} = a_k$$

$$R^{(1,l+k)}_{u+1-l}\ R^{(1,l-1+k)}_{u+1-l+1}\ R^{(1,l-2+k)}\ \cdots\ R^{(1,1+k)}\ R^{(2,l+k)}\ \cdots$$

$$\overrightarrow{\prod_{1\le i<j\le k}} R^{(k+i,k+j)} = \left(R^{(k+1,k+i+1)} \cdots R^{(k+1,k+l)}\right) \cdots \left(R^{k+l-1,k+l}_{(-1)}\right)$$

$$R^{k+l-2,k+l-1}\ R^{k+l-1,k+l}$$

$$\boxed{R(w) = u+P}$$

---

① $L_k = \overrightarrow{\prod_{1\le i\le k}} \overrightarrow{\prod_{i<j\le k}} R^{(i,j)}(i-j)$

$R_k = \overleftarrow{\prod_{1\le j\le k}} \overleftarrow{\prod_{1\le i<j}} R^{(ij)}_{(i-j)} = A_{[kj}(-1)^k \prod_{j=1}^{k}(-j)^{k-j+1}$

$k=3$ $\quad L_3 = R^{(12)}(-1)R^{(13)}(-2)R^{(23)}(-1)$

$R_3 = R^{(23)}(-1)R^{(13)}(-2)R^{(12)}(-1)$

$R^{(k-2,k-1)}\ R^{(k-2,k)}$

$L_{k-1} = R_{k-1}$

$L_k = \left[R^{(12)}(-1)\cdots R^{(1k)}(-k+1)\right]\left[R^{(23)}(-1)\cdots R^{(2k)}(-k+2)\right]\cdots\left[R^{(k-1,k)}(-1)\right]$

$= \left[R^{(12)}(-1)\cdots R^{(1,k-1)}(-k+2)\right] = L_{k-1}\left[R^{(1k)}\ R^{(2k)}\cdots R^{(k-1,k)}\right]$

$= \left[R^{(12)}\cdots R^{(1,k-1)}\right]$

$R^{(12)}\ R^{(13)}\ R^{(14)}\ R^{(23)}\ R^{(24)}\ R^{(34)}$

$R^{(12)}\ R^{(13)}\ R^{(23)}\ R^{(14)}\ R^{(24)}\ R^{(34)}$

$R^{(23)}\ R^{(13)}\ R^{(12)}\ R^{(14)}\ R^{(24)}\ R^{(34)}$

$R^{(23)}\ R^{(13)}\ R^{(24)}\ R^{(14)}\ R^{(12)}\ R^{(34)}$

$R^{(23)}\ R^{(24)}\ R^{(13)}\ R^{(14)}\ R^{(34)}\ R^{(12)}$

$R^{(23)}\ R^{(24)}\ R^{(34)}\ R^{(14)}\ R^{(13)}\ R^{(12)}$

$R^{(34)}\ R^{(24)}\ R^{(23)}\ R^{(14)}\ R^{(13)}\ R^{(12)}$

$R^{(34)}\ R^{(24)}\ R^{(14)}\ R^{(23)}\ R^{(13)}\ R^{(12)}$

6. $F_{K-1} \cdots F_2$

$\|$

$R_{K-1}\left[\ R^{(1K)}\ R^{(2K)} \cdots\cdots R^{(K-1,K)}\right]$

$\cdots R^{(12)}\ R^{(1K)}\ R^{(2K)} \cdots\cdots R^{(K-1,K)}$

$R^{(2K)}\ R^{(1K)}\ R^{(3K)} \cdots R^{(K-1,K)}\ R^{(12)}$

$F_K\ F_{K-1} \cdots F_4\ R^{(23)} R^{(13)}\ R^{(1K)}\ R^{(3K)} \cdots R^{(K-1,K)}\ F_2$

$R^{(23)}\ R^{(2K)} R^{(3K)}\ R^{(1K)}\ R^{(13)}\ R^{(4K)} \cdots R^{(K-1,K)} F_2$

6

$R^{(3K)} R^{(2K)}\ R^{(1K)}\ R^{(4K)} \cdots R^{(K-1,K)}\ R^{(23)}\ R^{(13)} F_3 F_2$

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$F_k \cdots F_{l+1}\ R^{lk}\ R^{l-1,k} \cdots R^{1k}\ R^{l+1,k} \cdots R^{(k-1,k)}\ F_l \cdots F_2$

$R^{l,l+1} \cdots R^{1,l+1}\ R^{lk}\ R^{l-1,k} \cdots R^{1k}\ R^{l+1,k} \cdots R^{(k-1,k)}$

$\left(R^{l,l+1} \cdots R^{2,l+1}\right)\left(R^{lk} \cdots R^{2k}\right) R^{1,l+1} \cdots R^{1k}\ R^{l+1,k} \cdots$

$R^{l+1,k}\ R^{1k}\ R^{1,l+1}\ R^{l+2,k} \cdots R^{(k-1,k)} \longrightarrow$

$l \le a \le 1$

$R^{a+1,l+1}\ R^{lk} \cdots R^{a+1,k}\ R^{l+1,k} \cdots R^{ak} \cdots R^{1k}\ R^{a,l+1} \cdots R^{1,l+1}$

$R^{lk} \cdots R^{a+2,k}\ R^{a+1,l+1}\ R^{a+1,k}\ R^{l+1,k}\ R^{ak} \cdots R^{1k}$

$R^{lk} \cdots R^{a+2,k}\ R^{l+1,k}\ R^{a+1,k}\ R^{a+1,l+1}\ R^{ak} \cdots R^{1k}$

$R^{k-1,k}\ R^{k-2,k} \cdots R^{1,k}\ F_{k-1} \cdots F_2$

$LH_k = RH_k$

② $A_{\{K\}}^{(1,\cdots,K)} A_{\{l\}}^{(K+1,\cdots,K+l)} \overrightarrow{\prod_{1\le i\le K}} \overleftarrow{\prod_{1\le j\le l}} R^{(i,j+K)}(u+i-j-k+l)$

$= \left(\overleftarrow{\prod_{1\le i\le K}} \overrightarrow{\prod_{1\le j\le l}} R^{(i,j+K)}(u+i-j-k+l) \right) A_{\{K\}}^{(1,\cdots,K)} A_{\{l\}}^{(K+1,\cdots,K+l)}$

$F_1 F_2 \cdots F_{l-1} \left(R^{(1,K+l)} R^{(1,K+l-1)} \cdots R^{(1,K+1)} \right)\left(R^{(2,K+l)} R^{(2,K+l-1)} \cdots R^{(2,K+1)} \right) \cdots ( )$

$\qquad\qquad\qquad

$$LHS = A_{\{k\}}^{(1,\dots,k)} \overrightarrow{\prod_{1\le i\le k}} \overrightarrow{\prod_{1\le j\le l}} R^{(i,k+j)}(u+i-j-k+l)\, A_{\{l\}}^{(k+1,\dots,k+l)}$$

$$F_1 \cdots F_{k-1} \left(R^{1,k+1}\cdots R^{1,k+l}\right)\cdots\left(R^{k,k+1}\cdots R^{k,k+l}\right)$$

$$G_1 \cdots G_k$$

$$F_i = R^{i,i+1}\cdots R^{i,k}$$

$$R^{k-1,k} G_i = G_i R^{k-1,k} \quad \text{for } 1\le i\le k-2$$

$$R^{k-1,k} R^{k-1,k+1} R^{k,k+1}\left(R^{k-1,k+2}\cdots R^{k-1,k+l}\right)\left(R^{k,k+2}\cdots R^{k,k+l}\right)$$

$$\left(R^{k,k+1}\right)\left(R^{k-1,k+1}\right) R^{k-1,k} R^{k-1,k+2} R^{k,k+2}\left(R^{k-1,k+3}\cdots R^{k-1,k+l}\right)\left(R^{k,k+3}\cdots R^{k,k+l}\right)$$

$$R^{k,k+1} R^{k-1,k+1} R^{k,k+2} R^{k-1,k+2} R^{k-1,k}(\quad)(\quad)$$

$$R^{k,k+1} R^{k,k+2} R^{k-1,k+1} R^{k-1,k+2} R^{k-1,k}(\quad)(\quad)$$

$$\vdots$$

$$F_{k-1} G_1 \cdots G_k = G_1 \cdots G_{k-2} G_k G_{k-1} R^{k-1,k} F_{k-1}$$

$$F_{k-2} = R^{k-2,k-1} R^{k-2,k}$$

$$F_{k-2} F_{k-1} G_1 \cdots G_k = F_{k-2} G_1 \cdots G_{k-2} G_k G_{k-1} F_{k-1}$$

$$= G_1 \cdots G_{k-3} F_{k-2} G_{k-2} G_k G_{k-1} F_{k-1} = G_1 \cdots G_{k-3} G_k G_{k-1} G_{k-2} F_{k-2} F_{k-1}$$

$$R^{k-2,k-1} R^{k-2,k} G_{k-2} G_k G_{k-1} = R^{k-2,k-1} G_k G_{k-2} R^{k-2,k} G_{k-1}$$

$$= G_k R^{k-2,k-1} G_{k-2} G_{k-1} R^{k-2,k} = G_k G_{k-1} G_{k-2} R^{k-2,k-1} R^{k-2,k}$$

(2R)

$$R^{\Lambda_K,\Lambda_\ell}(u) = \overleftarrow{\prod_{1\le i\le K}} \overrightarrow{\prod_{1\le j\le \ell}} R^{i,j+K}(u+i-j-K+\ell)\, A_{\{K\}}^{(1,\dots,K)}\, A_{\{\ell\}}^{(K+1,\dots,K+\ell)}$$

(3)

$$\overleftarrow{\prod_{1\le i\le K}} \overrightarrow{\prod_{1\le j\le \ell}} R^{i,K+j}(u-v+i-j-K+\ell)\, A_{\{K\}}^{1,\dots,K}\, A_{\{\ell\}}^{K+1,\dots,K+\ell} \cdot \overleftarrow{\prod_{K+1\le i\le K}} \overrightarrow{\prod_{1\le j\le m}} R^{i,K+\ell+j}(u+i-j-K+m)\, A_{\{K\}}^{1,\dots,K}\, A_{\{m\}}^{K+\ell+1,\dots,K+\ell+m}$$

$$\cdot \overleftarrow{\prod_{1\le i\le \ell}} \overrightarrow{\prod_{1\le j\le m}} R^{K+i,K+\ell+j}(v+i-j-\ell+m)\, A_{\{\ell\}}^{K+1,\dots,K+\ell}\, A_{\{m\}}^{K+\ell+1,\dots,K+\ell+m}$$

$$\overleftarrow{\prod_{1\le i\le K}} \overrightarrow{\prod_{1\le j\le \ell}} R^{i,K+j}(u-v+i-j-K+\ell) \overrightarrow{\prod_{1\le i\le K}} \overleftarrow{\prod_{1\le j\le m}} R^{i,K+\ell+j}(u+i-j-K-m)$$

$$A_K A_\ell \overrightarrow{\prod_{1\le i\le K}} \overleftarrow{\prod_{1\le j\le \ell}} R^{(i,K+j)}_{u-v+i-j-K+\ell} \overleftarrow{\prod} \overrightarrow{\prod} R^{(i,K+\ell+j)}_{u+i-j-K+m} \overleftarrow{\prod}$$

$$RHS = \overleftarrow{\prod}\overrightarrow{\prod} A_\ell A_m \overleftarrow{\prod}\overrightarrow{\prod} A_K A_m\, A_K A_\ell \overrightarrow{\prod}\overleftarrow{\prod}$$

$$\frac{1}{2}(v_1\otimes v_2 - v_2\otimes v_1) \quad \overleftarrow{\prod}\overrightarrow{\prod} A_\ell A_m \overleftarrow{\prod}\overrightarrow{\prod} A_K A_m\, A_\ell \overrightarrow{\prod}\overleftarrow{\prod}$$

$$\frac{1}{2}\left[\frac{1}{2}(v_1\otimes v_2 - v_2\otimes v_1) - \frac{1}{2}(v_2\otimes v_1 - v_1\otimes v_2)\right]$$

$$= \frac{1}{2}(v_1\otimes v_2 - v_2\otimes v_1)$$

$$\overleftarrow{\prod}\overrightarrow{\prod} A_\ell A_m A_K \overrightarrow{\prod}\overrightarrow{\prod} A_m A_\ell \overrightarrow{\prod}\overleftarrow{\prod}$$

$$R^{K-1,K} R^{K,K+1} R^{K-1,K+1}$$

$$A_K A_\ell A_m \underbrace{\left(\overleftarrow{\prod}\overleftarrow{\prod}\right)}_{23} \underbrace{\left(\overrightarrow{\prod}\overrightarrow{\prod}\right)}_{13} \underbrace{\overleftarrow{\prod}\overleftarrow{\prod}}_{12} A_m A_\ell$$

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$$\overleftarrow{\prod}\overrightarrow{\prod} R^{i,k+j}\ \overrightarrow{\prod}\overrightarrow{\prod} R^{i,k+l+j}\ \overrightarrow{\prod}\overleftarrow{\prod} R^{k+i,k+l+j} \qquad A_{kj}\ A_{lj}\ A_{mj}$$

$F_k \cdots F_1 \qquad G_1 \cdots G_k \qquad H_1 \cdots H_l$

$F_i = R^{i,k+1} \cdots R^{i,k+l}$, $G_i = R^{i,k+l+1} \cdots R^{i,k+l+m}$, $H_i = R^{k+i,k+l+m} \cdots R^{k+i,k+l+1}$

$$A_k A_l A_m \overrightarrow{\prod}\overleftarrow{\prod} R^{i,k+j}\ \overrightarrow{\prod}\overleftarrow{\prod} R^{i,k+l+j}\ \overrightarrow{\prod}\overleftarrow{\prod} R^{k+i,k+l+j}$$

$F_1 \cdots F_k \qquad G_1 \cdots G_k \qquad H_1 \cdots H_l$

$F_i = R^{i,k+l} \cdots R^{i,k+1}$, $G_i = R^{i,k+l+m} \cdots R^{i,k+l+1}$, $H_i = R^{k+i,k+l+m} \cdots R^{k+i,k+l+1}$

~~$R^{k,k+1}$ $R^{l,k+l+m}$~~

$F_1 \cdots F_{k-1}\ G_1 \cdots G_{k-1}\ F_k G_k\ H_1 \cdots H_l$

$F_k G_k H_1 \cdots H_l = R^{k,k+l} \cdots R^{k,k+1}\ R^{k,k+l+m} \cdots R^{k,k+l+1}\ R^{k+1,k+l+m} \cdots R^{k+1,k+l+1}$ $H_2 \cdots H_l$

$= R^{k,k+l} \cdots R^{k,k+2}\ H_1 G_k\ R^{k,k+1}_{H_2 \cdots H_l} = H_1 R^{k,k+l} \cdots R^{k,k+2} G_k H_2 \cdots H_l R^{k,k+1}$

$= H_1 \cdots H_l\ G_k F_k$

$F_1 \cdots F_k\ G_1 \cdots G_k\ H_1 \cdots H_l = H_1 \cdots H_l\ G_1 \cdots G_k\ F_1 \cdots F_k$

$$\prod_{1\le i\le k}^{\rightarrow}\ \prod_{1\le j\le l}^{\leftarrow} R^{(i,k+j)}(u+i-j-k+l) \qquad \prod_{1\le j\le k}^{\leftarrow}\ \prod_{1\le i\le l}^{\rightarrow} R^{(j,k+i)}((u+j-i-l+k)(-1))$$

$F_1 \cdots F_k\ G_k \cdots G_1$ where $F_i = R^{(i,k+l)}(u+i-k-l+k) \cdots R^{(i,k+1)}(u+i-k-l+k)$

$R^{(k,k+1)}(u+k-k-1-k+l)\left(R^{(k,k+1)}(-u-k+k+1+k-l)\right)$ $G_i = R^{(i,k+1)} \cdots R^{(i,k+l)}$

(4) $R^{\Lambda^K,\Lambda^\ell}(u)\left(R^{\Lambda^\ell,\Lambda^K}(-u)\right)^{(21)} = \prod_{1\le i\le K}\prod_{1\le j\le \ell}\left(1-(u-i+j)^2\right)$

← interchanges K and ℓ everywhere

Proof :—

$$LHS = A^{(1)}_{[K]}A^{(2)}_{[\ell]}\overrightarrow{\prod_{1\le i\le K}}\overleftarrow{\prod_{1\le j\le \ell}}R^{(i,K+j)}(u+i-j-K+\ell)\overleftarrow{\prod_{1\le i\le K}}\overrightarrow{\prod_{1\le j\le \ell}}R^{(K+j,i)}(-u-i+j+K-\ell)A^{(1)}_{[K]}A^{(2)}_{[\ell]}$$

$$= F_1 \cdots F_K \; G_K \cdots G_1$$

$$F_i = R^{(i,K+\ell)}(u+i-j-K+\ell)\cdots R^{(i,K+1)}(u+i-j-K+\ell)$$

$$G_i = R^{(K+\ell,i)}(-u-i+j+K-\ell)\cdots R^{(K+\ell,i)}(u-i+j+K-\ell)$$

$$\prod_{i=1}^{K}\prod_{j=1}^{\ell}\left(1-(u+i-j-K+\ell)^2\right) = \prod_{i=1}^{\ell}\prod_{j=1}^{\ell}\left(1-(u-i+j)^2\right)$$

$$u+i-j-K+\ell = u-(K+1-i)+(\ell+1-j)$$

$$-1 \quad +1$$

(5) For any $Q \in End(V)$,

$$\left[\underbrace{R^{\Lambda^K,\Lambda^\ell}(u)}_{\uparrow},\ Q^{\Lambda^K}\otimes Q^{\Lambda^\ell}\right] = 0$$

$$= A^{(1)}_{[K]}A^{(2)}_{[\ell]}\,Q^{\otimes K}\otimes Q^{\otimes \ell}$$

Proof: each term in this product commutes with $Q^{\otimes K}\otimes Q^{\otimes \ell}$

since $\left[R(u),\ Q\otimes Q\right] = 0$

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$$A^{(1,-K)}_{\{K\}}\, A^{(K+1,\infty)}_{\{K\}} \prod_{1\le i\le K}\ \prod_{1\le j\le \ell} R^{(i,K+j)}(u+i-j-K+\ell) \Big/ \left(\prod_{1\le i\le \ell}\ \prod_{1\le j\le K} R^{(i,\ell+j)}(-u+i+j-K+\ell) \right)^{(21)} A^{(1,\infty)}_{\{\ell\}}\, A^{(1,\infty)}_{\{K\}}$$

$$R^{(K,K+1)}(u-1-K+\ell) \Big/ \left(R^{(\ell,\ell+1)}(-u+1-K+\ell) \right)^{(21)}$$

$$Y_N \equiv Y(\mathfrak{gl}_N) = \left\langle T^{(s)}_{ab} \;;\; a,b = 1,\dots,N,\ s = 1,2,3,\dots \right\rangle$$

modulo

(6) RTT

$$R^{(12)}(u-v)\, T^{(13)}(u) T^{(23)}(v) = T^{(23)}(v) T^{(13)}(u) R^{(12)}(u-v) \in \mathrm{End}\left(\mathbb{C}^N_V \otimes \mathbb{C}^N_V \otimes Y_N \right)[[u]]$$

where

$$T(u) = \sum_{a,b=1}^{N} E_{ab} \otimes T_{ab}(u) \in \mathrm{End}\left(\mathbb{C}^N \right) \otimes (Y_N)[[u^{-1}]]$$

and $\quad T_{ab}(u) = \delta_{ab} + \sum_{s=1}^{\infty} T^{(s)}_{ab} u^{-s} \in Y_N[[u^{-1}]]$.

---

(7) For $K \ge 1$, consider $T^{(K,K+1)}(u) \cdots\cdots T^{(1,K+1)}_{(u-K+1)} \in \mathrm{End}(V^{\otimes K}) \otimes Y_N[[u]]$

we have

$$A_{\{K\}}\, T^{(1,K+1)}(u-K+1) \cdots T^{(K,K+1)}(u) = T^{(K,K+1)}(u) \cdots T^{(1,K+1)}(u-K+1)\, A_{\{K\}}$$

Proof :-

Claim implies (7) because of (1)

$$\overleftarrow{\prod_{1\le i<j\le K}} R^{(ij)}(i-j)\, T^{(1,K+1)}(u-K+1) \cdots T^{(K,K+1)}(u) = T^{(K,K+1)}(u) \cdots T^{(1,K+1)}(u-K+1) \overleftarrow{\prod_{1\le i<j\le K}} R^{(ij)}(i-j)$$

($i-j$: in general $u_i - u_j$; $u_1 \cdots u_K$; u_K; u_1; $u_i - u_j$)

So, let $R^{(ij)}(u_i - u_j)$ to be R^{ij} and $T^{(i,K+1)}_{(u_i)}$ to be T^i

Want to check

$$\left(\overleftarrow{\prod_{1\le i<j\le K}} R^{ij}\right)(T_1 T_2 \cdots T_K) = (T_K T_2 \cdots T_1)\left(\overleftarrow{\prod_{1\le i<j\le K}} R^{ij}\right)$$

(1) Observe $(R^{1K} \cdots R^{12}) T_1 (T_2 \cdots T_K) = (T_2 \cdots T_K) T_1 (R^{1K} \cdots R^{12})$

$$(R^{1K} \cdots R^{13})(R^{12} T_1 T_2)(T_3 \cdots T_K)$$

$$(R^{1K} \cdots R^{13})(T_2 T_1 R^{12})(T_3 \cdots T_K) = T_2 (R^{1K} \cdots R^{13} T_1 T_3 \cdots T_K) R^{12}$$

T_3 moves left and R^{13} moves right

Then, $\left(\overleftarrow{\prod_{1\le i<j\le K}} R^{ij}\right)(T_1 \cdots T_K) = \left(\overleftarrow{\prod_{2\le i<j\le K}} R^{ij}\right)(T_2 \cdots T_K) T_1 (R^{1K} \cdots R^{12})$ — use observation

Now interchange T_2 with $T_3 \cdots T_K$ ▨

---

(2) Define (T) $T^{\wedge K}(u) = T^{(K,K+1)}(u) \cdots T^{(1,K+1)}(u-K+1) \Big|_{V^{\wedge K}}^{(1,\cdots,K)}$

$\in \mathrm{End}(V^{\wedge K}) \otimes Y_N[[u^{-1}]]$

In particular, $T^{\wedge 1}(u) = T(u)$.

---

(3) $\left(R^{\wedge K,\wedge \ell}(u-v)\right)^{(12)} \left(T^{\wedge K}(u)\right)^{(13)} \left(T^{\wedge \ell}(v)\right)^{(23)}$

$\in \mathrm{End}(V^{\wedge K} \otimes V^{\wedge \ell}) \otimes Y_N[[u,v]]$ (1, 2, 3)

$= \left(T^{\wedge \ell}(v)\right)^{(23)} \left(T^{\wedge K}(u)\right)^{(13)} \left(R^{\wedge K,\wedge \ell}(u-v)\right)^{(12)}$

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Proof :-

$$LHS = A^{(1)}_{\{K\}} A^{(2)}_{\{\ell\}} \overrightarrow{\prod_{1\le i\le K}} \overleftarrow{\prod_{1\le j\le \ell}} R^{(i,K+j)}_{(u+i-j-K+\ell)} T^{(i,K+\ell+1)}_{(u-K+i)} \cdots T^{(K,K+\ell+1)}_{(u)} T^{(K+1)}_{(v-\ell+1)} \cdots T^{(K+\ell)}_{(v)}$$

$F_1 F_2 \cdots F_K$; $u+i-K$, $i: 1 \to K$; $v+j-\ell$, $j: 1 \to \ell$

$$F_i = R^{(i,K+\ell)}_{(u-v+i-\ell-K+\ell)} \cdots R^{(i,K+1)}_{(u-v+i-1-K+\ell)}$$

$$F_K T^{(1)}_{(u-K+1)} \cdots T^{(K)}_{(u)} T^{(K+1)}_{(v-\ell+1)} \cdots T^{(K+\ell)}_{(v)} = R^{(K,K+\ell)} \cdots R^{(K,K+2)} R^{(K,K+1)}_{(u-v+\ell-1-K+\ell)} T^{(1)}_{(u-K+1)} \cdots T^{(K)}_{(u)} T^{(K+1)}_{(v-\ell+1)} \cdots T^{(K+\ell)}_{(v)}$$

$$= R^{(K,K+\ell)} \cdots R^{K,K+2} T^{(1)} \cdots T^{K-1} R^{K,K+1} T^{K} T^{(K+1)} T^{K+2} \cdots T^{K+\ell}$$

$$= \left(R^{K,K+\ell} \cdots R^{K,K+2}\right)\left(T^{1} \cdots T^{K-1}\right)\left(T^{K+1} T^{K}\right)\left(T^{K+2} \cdots T^{K+\ell}\right) R^{K,K+1}$$

$$= \cdots = \left(T^{1} \cdots T^{K-1}\right)\left(T^{K+1} \cdots T^{K+\ell}\right) T_K F_K$$

So, $F_K (T^1 \cdots T^K)(T^{K+1} \cdots T^{K+\ell}) = (T^1 \cdots T^{K-1})(T^{K+1} \cdots T^{K+\ell}) T^K F_K$

$$(F_1 \cdots F_K)(T^1 \cdots T^K)(T^{K+1} \cdots T^{K+\ell}) \cdots = (T^{K+1} \cdots T^{K+\ell})(T^1 \cdots T^K)(F_1 \cdots F_K)$$

$= RHS$

using (2) now to move $A^{(1)}_{\{K\}} A^{(2)}_{\{\ell\}}$.

---

(9) There exists $qdet\, T(u) = 1 + \sum_{s=1}^{\infty} d_s u^{-s} \in Y_N[[u^{-1}]]$

given by $T^{\wedge N}(u) = \underbrace{qdet\, T(u)}_{\text{quantum determinant}} A_{\{N\}}$

Proof :- $\dim V^{\wedge N} = 1$, $A_{\{N\}}$ is 1-dim proj

Note that each $T^{(i,K+1)}(u-K+i)$ starts with 1.

(10) we have

$$q\det T(u) = (-1)^{\sigma} \sum_{\tau \in S_N} (-1)^{\tau} T^{(u)}_{\sigma(N),\tau(N)} \cdots T^{(u-N+1)}_{\sigma(1),\tau(1)}$$

$$= (-1)^{\sigma} \sum_{\tau \in S_N} (-1)^{\tau} T^{(u-N+1)}_{\tau(1),\sigma(1)} \cdots T^{(u)}_{\tau(N),\sigma(N)}$$

Proof :— where $\sigma \in S_N$ is arbitrary.

$$q\det T(u) A_{\{N\}} = T^{(N,N+1)}(u) \cdots T^{(1,N+1)}(u-N+1) A_{\{N\}}$$

$$= A_{\{N\}} T^{(1,N+1)}_{(u-N+1)} \cdots T^{(N,N+1)}(u)$$

Apply both sides to $e_1 \otimes \cdots \otimes e_N$.

$$LHS = q\det T(u)(e_1 \wedge \cdots \wedge e_N)$$

$$RHS1 = \sum_{\substack{i_1,\cdots,i_N \\ j_1,\cdots,j_N=1}}^{N} T_{i_N j_N}(u) \cdots T_{i_1 j_1}(u-N+1)\left(E_{i_1 j_1} \otimes \cdots \otimes E_{i_N j_N}\right) e_1 \wedge \cdots \wedge e_N$$

$$= \sum_{i_1,\cdots,i_N=1}^{N} \sum_{\tau \in S_N} (-1)^{\tau} T_{i_N,\tau(N)}(u) \cdots T_{i_1,\tau(1)}(u-N+1)\, e_{i_1} \otimes \cdots \otimes e_{i_N}$$

coeff. of $e_1 \wedge \cdots \wedge e_N \rightarrow \sum_{\tau \in S_N} (-1)^{\tau} T_{N,\tau(N)}(u) \cdots T_{1,\tau(1)}(u-N+1)$

$$RHS2 = A_{\{N\}} \sum_{\substack{i_1,\cdots,i_N \\ j_1,\cdots,j_N=1}}^{N} T_{i_1 j_1}(u-N+1) \cdots T_{i_N j_N}(u)\left(E_{i_1 j_1} \otimes \cdots \otimes E_{i_N j_N}\right) e_1 \otimes \cdots \otimes e_N$$

$$= \sum_{i_1,\cdots,i_N=1}^{N} T_{i_1,1}(u-N+1) \cdots T_{i_N,N}(u) \underbrace{A_{\{N\}}(e_{i_1} \otimes \cdots \otimes e_{i_N})}_{e_{i_1} \wedge \cdots \wedge e_{i_N}}$$

$$= \sum_{\tau \in S_N} (-1)^{\tau} T_{\tau(1),1}(u-N+1) \cdots T_{\tau(N),N}(u)(e_1 \wedge \cdots \wedge e_N)$$

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(11) The coefficients of $q\det T(u)$ are free generators of the center of Y_N.

(12) $tr_W : End(W) \to \mathbb{C}$

For any $Q \in End(V)$ define the series

$$\tau_{k,Q}(u) = (tr_{V^{\wedge k}} \otimes id)\left(Q^{\wedge k} T^{\wedge k}(u)\right) \in Y_N[[u^{-1}]]$$

for $k = 1, \dots, N$.

and

$\tau_{0,Q} = 1$. (Transfer matrices)

(*) $\tau_{N,Q} = \det Q \; q\det T(u)$

(*) $[\tau_{k,Q}(u), \tau_{l,Q}(v)] = 0$

$$\tau_{k,Q}(u)\tau_{l,Q}(v) = \left(tr_{V^{\wedge k}} \otimes tr_{V^{\wedge l}} \otimes id\right)\left(\left(Q^{\wedge k} T^{\wedge k}(u)\right)^{(13)}\left(Q^{\wedge l} T^{\wedge l}(v)\right)^{(23)}\right)$$

$$= \left(tr_{V^{\wedge k}} \otimes tr_{V^{\wedge l}} \otimes id\right)\left(\left(Q^{\wedge k}\right)^{(1)}\left(Q^{\wedge l}\right)^{(2)}\left(T^{\wedge k}(u)\right)^{(13)}\left(T^{\wedge l}(v)\right)^{(23)}\right)$$

$$= \left(tr_{V^{\wedge k}} \otimes tr_{V^{\wedge l}} \otimes id\right)\left[\left(Q^{\wedge k}\right)^{(1)}\left(Q^{\wedge l}\right)^{(2)}\left(R^{\wedge k, \wedge l}(u-v)\right)^{(12)-1}\left(T^{\wedge l}(v)\right)^{(23)}\left(T^{\wedge k}(u)\right)^{(13)}\left(R^{\wedge k, \wedge l}(u-v)\right)^{(12)}\right]$$

$$= \left(tr_{V^{\wedge k}} \otimes tr_{V^{\wedge l}} \otimes id\right)\left[\left(Q^{\wedge l} T^{\wedge l}(v)\right)^{(23)}\left(Q^{\wedge k} T^{\wedge k}(u)\right)^{(13)}\right]$$

$$= \tau_{l,Q}(v)\,\tau_{k,Q}(u)$$ ∎

(*) If Q has a simple spectrum, then the coeff. of the series $\tau_{1,Q}(u), \dots, \tau_{N-1,Q}(u)$ and $q\det T(u)$ generate a maximal commutative subalgebra in Y_N.

(11a) $q\det T(u)$ lies in the centre of $\mathcal{Y}_N$, that is, d_i are central

Proof :—

Consider $V^{\otimes} \otimes \underbrace{V \otimes \cdots \otimes V}_{N \text{ copies}} = V^{\otimes(N+1)}$ (0, 1, ---, N)

Let $T_0(v) = T^{(0,N+1)}_{(v)}$, $T_1 = T^{(1,N+1)}_{(u-N+1)}$, ---, $T_N = T^{(N,N+1)}_{(u)}$

Claim $\boxed{T^{(0,N+1)}_{(v)}\, q\det T(u)\, A_N = q\det T(u)\, T_0(v)\, A_N}$

$$\overleftarrow{\prod_{0\le i<j\le N}} R^{(ij)}(u_i-u_j)\, T^{(0,N+1)}_{(u_0)} \cdots T^{(N,N+1)}_{(u_N)} = T^{(N,N+1)}_{(u_N)} \cdots T^{(0,N+1)}_{(u_0)} \overleftarrow{\prod_{0\le i<j\le N}} R^{(ij)}(u_i-u_j) \quad (*)$$

$u_0 = v$, $u_i = u-N+i$, $1\le i\le N$.

$$A_{\{N\}} \overleftarrow{\prod_{1\le j\le N}} R^{(0j)}(v-u+N-j)\, T_0 T_1 \cdots T_N = T_N \cdots T_1 T_0\, A_{\{N\}} \overleftarrow{\prod_{1\le j\le N}} R^{(0j)}(v-u+N-j) \quad (a)$$

Subclaim

$$A_{\{N\}} R^{0N} \cdots R^{01} = R^{01} \cdots R^{0N} A_{\{N\}} \quad (b)$$

$$\Rightarrow A_{\{N\}} (R^{01})^{-1} \cdots (R^{0N})^{-1} = (R^{0N})^{-1} \cdots (R^{01})^{-1} A_{\{N\}}$$

Same proof since $R^{ij}(R^{0i})^{-1}(R^{0j})^{-1} = (R^{0j})^{-1}(R^{0i})^{-1} R^{ij}$

$$R^{i0} R^{j0} R^{ij} = R^{ij} R^{j0} R^{i0}$$

LHS of (a) $= A_{\{N\}} R^{0N} \cdots R^{01} T_0 \cdots T_N$

$= A^2_{\{N\}} R^{0N} \cdots R^{01} T_0 \cdots T_N$

$= A_{\{N\}} R^{01} \cdots R^{0N} A_{\{N\}} T_0 \cdots T_N$ ↘ (b)

$= \underbrace{\left(A_{\{N\}} R^{01} \cdots R^{0N} A_{\{N\}}\right)}_{f(u,v) A_{\{N\}}} T_0 \left(q\det T(u) A_{\{N\}}\right)$

RHS of (a) $= \left(T_N \cdots T_1 A_{\{N\}}\right) T_0 A_{\{N\}} R^{0N} \cdots R^{01}$

↘ (b)

$= \left(q \det T(u) A_{\{N\}}\right) T_0 \underbrace{\left(A_{\{N\}} R^{01} \cdots R^{0N} A_{\{N\}}\right)}_{f(u,v) A_{\{N\}}}$

Subclaim $A_{\{N\}} R^{(01)}_{(v-u+N-1)} \cdots R^{(0N)}_{(v-u)} A_{\{N\}} = f(u,v) A_N$ with $f(u,v) \neq 0$

↓

acts on $V^{\otimes(N+1)} = V \otimes V^{\otimes N}$ as $X \otimes A_{\{N\}}$

$A_{\{N\}} V^{\otimes(N+1)} \longmapsto V \otimes \mathbb{C}$

Action of $GL_N(\mathbb{C})$ on $V^{\otimes(N+1)}$ as $Q \cdot (v_0 \otimes \cdots \otimes v_N) = Qv_0 \otimes \cdots \otimes Qv_N$

But $R^{(0i)}$ commute with $Q \otimes Q$

$\Rightarrow X$ is a scalar operator.

Clearly not zero $\rightarrow (v-u) \cdots (v-u+N-1)$

↑ top degree term

Finally, $T_0(v)\, q\det T(u) A_{\{N\}} = q\det T(u)\, T_0(v) A_{\{N\}}$

(11b) The coefficients $d_1, d_2, \ldots$ of $\mathrm{qdet}\, T(u)$ are algebraically independent and generate the whole centre of Y_N.

Define a grading on Y_N by :-

$$\deg T_{ab}^{\{s\}} = s-1$$

Let $\mathrm{gr}\, Y_N$ denote the corresponding graded algebra

(13) $\mathrm{gr}\, Y_N \cong U[\mathfrak{gl}_N[x]] = U(\mathfrak{gl}_N \otimes \mathbb{C}[x]) = \langle E_{ij} x^{s-1} \rangle$, $1 \le i, j \le N$, $s = 1, 2, \ldots$

Proof :-

$$RTT \Leftrightarrow (u-v)[T_{ab}(u), T_{cd}(v)] = T_{cb}(v) T_{ad}(u) - T_{cb}(u) T_{ad}(v)$$

$$\| $$

$$(u-v+P) \sum_{a,b,c,d} T_{ab}(u) T_{cd}(v) (E_{ab} \otimes E_{cd})$$

$$= \sum_{a,b,c,d} T_{cd}(v) T_{ab}(u) (E_{ab} \otimes E_{cd}) [(u-v)+P]$$

$$\Leftrightarrow \sum_{a,b,c,d} (u-v)[T_{ab}(u), T_{cd}(v)] (E_{ab} \otimes E_{cd})$$

$$= \sum_{a,b,c,d} T_{ab}(u) T_{cd}(v) P (E_{ab} \otimes E_{cd}) - T_{cd}(v) T_{ab}(v) (E_{ab} \otimes E_{cd}) P$$

$E_{cb} \otimes E_{ad}$ $\quad$ $E_{ad} \otimes E_{cb}$

$$(u-v)[T_{ab}(u), T_{cd}(v)] = T_{cb}(v) T_{ad}(u) - T_{cb}(u) T_{ad}(v)$$

$$\Updownarrow$$

$$[T_{ab}^{\{s_1\}}, T_{cd}^{\{s_2\}}] = \sum_{r=0}^{\min(s_1,s_2)-1} T_{cb}^{\{r\}} T_{ad}^{\{s_1+s_2-1-r\}} - T_{cb}^{\{s_1+s_2-1-r\}} T_{ad}^{\{r\}}$$

Here $T_{ab}^{\{0\}} = \delta_{ab}$

degree on LHS $= s_1+s_2-2$

degree on RHS s_1+s_2-3 except when $r=0$

$r=0$ term on RHS is :-

$$T_{cb}^{\{0\}} T_{ad}^{\{s_1+s_2-1\}} - T_{cb}^{\{s_1+s_2-1\}} T_{ad}^{\{0\}} = \delta_{cb} T_{ad}^{\{s_1+s_2-1\}} - \delta_{ad} T_{cb}^{\{s_1+s_2-1\}}$$

$\Rightarrow$ In $\mathrm{Fr}Y_N$ relations are

$$\left[\hookrightarrow T_{ab}^{\{s_1\}}, \hookrightarrow T_{cd}^{\{s_2\}} \right] = \delta_{cb} \hookrightarrow T_{ad}^{\{s_1+s_2-1\}} - \delta_{ad} \hookrightarrow T_{cb}^{\{s_1+s_2-1\}}$$

where $\hookrightarrow T_{ab}^{\{s\}}$ is the image of $T_{ab}^{\{s\}}$ in $(s-1)^{th}$ graded component

identifying $\hookrightarrow T_{ab}^{\{s\}}$ and $E_{ab} x^{s-1}$

we have homomorphism $U(\mathfrak{gl}_N[x]) \longrightarrow \mathrm{Fr}Y_N$ onto

$\circledast$ ker is trivial ($\hookrightarrow T_{ab}^{\{s\}}$ are algebraically independent)

Proof of 11b :-

Let $Z = E_{11} + \cdots + E_{NN}$ so that $\mathfrak{gl}_N = \mathbb{C}Z \oplus \mathfrak{sl}_N$.

Claim: For any $s = 1, 2, 3, \cdots$, d_s has degree $s-1$ and its image in $(s-1)$-th component of $\mathrm{Fr}Y_N$ coincides with $Z x^{s-1}$

d_s is linear combination of monomials of the form

$$T_{\tau(1),1}^{\{s_1\}} \cdots T_{\tau(N),N}^{\{s_N\}} \quad \text{where } s_1+\cdots+s_N \le s$$

$\Rightarrow \deg(d_s) \le s-1$ with $\deg(d_s) = s-1$ when for some i we have $s_j = \delta_{ij} s$, $1 \le j \le N$

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In this case, since $T_{ab}^{\{0\}} = \delta_{ab}$, τ has to be trivial since otherwise $T_{P(1),1}^{\{s\}} = \delta_{P(1),1} = 0$

$$\Rightarrow d_s = T_{11}^{\{s\}} + \cdots + T_{NN}^{\{s\}} + (\text{terms of degree} < s-1)$$

Thus, $\deg(d_s) = s-1$ and in $\mathcal{F}r\mathcal{V}_N$, d_s is identified with zx^{s-1}

moreover, it follows that d_s, $s = 1, 2, \ldots$, are algebraically independent

Claim ⊖ Centre of $U(\mathcal{F}l_N[x]) = \langle z, zx, zx^2, \ldots \rangle$

$$U(\mathcal{F}l_N[x]) = \mathbb{C}[z, zx, zx^2, \ldots] \otimes U(sl_N[x])$$

~~$(U(\mathcal{F}_1 \oplus \mathcal{F}_2) = U(\mathcal{F}_1) \otimes U(\mathcal{F}_2))$~~

But centre of $U(sl_N[x])$ is trivial

□

(14) Let $\mathcal{F}$ be a Lie algebra whose centre is trivial. Then the centre of $U(\mathcal{F}[x])$ is also trivial.

Proof :— As $\mathcal{F}$-module $S(\mathcal{F}) \xrightarrow{\sim} U(\mathcal{F})$ is an isomorphism

To Prove : $S(\mathcal{F}[x])$ as adjoint $\mathcal{F}[x]$-module has no nontrivial invariant element

Let $\{e_1, \ldots, e_n\}$ be a basis of $\mathcal{F}$, and

$$[e_i, e_s] = \sum_{k=1}^{n} c_{ij}^{k} e_k \,, \quad c_{ij}^{k} \in \mathbb{C}$$

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Then $\prod_{i,r} (e_i x^r)$ (finite product)

form a basis of $S(\mathcal{F}[x])$.

Let $A \in S(\mathcal{F}[x])$ be a $\mathcal{F}[x]$-invariant element and m be a maximal integer s.t. $e_i x^m$ occurs in A for some $i = 1, \ldots, n$.

Then $A = \sum_d A_d (e_1 x^m)^{d_1} \cdots (e_n x^m)^{d_n}$

with $d = (d_1, \ldots, d_n)$, $d_i \geq 0$ and $A_d \neq$ a polynomial in variables $e_i x^r$ with $r < m$

A is $\mathcal{F}[x]$-invariant $\Rightarrow$ $ad e_i x (A) = 0$ for $i = 1, \ldots, n$.

Component containing elements of the form $e_k x^{m+1}$ must be zero.

that is,

$$\sum_d A_d \sum_{j=1}^n d_j (e_1 x^m)^{d_1} \cdots (e_j x^m)^{d_j - 1} \cdots (e_n x^m)^{d_n} \sum_{k=1}^n c_{ij}^k e_k x^{m+1} = 0$$

Taking coeff of $e_k x^{m+1}$:—

$$\sum_d A_d \sum_{j=1}^n d_j c_{ij}^k (e_1 x^m)^{d_1} \cdots (e_j x^m)^{d_j - 1} \cdots (e_n x^m)^{d_n} = 0$$

$$\Rightarrow \sum_{d'} A_{d' + \delta_j} (d'_j + 1) c_{ij}^k = 0, \quad 1 \leq i, k \leq n, \quad d' = (d'_1, \ldots, d'_n) \geq 0$$

For a fixed d', $e'_j = (d'_j + 1) e_j$, $j = 1, \ldots, n$ also form basis of $\mathcal{F}$.

$\mathcal{G}$ has trivial centre $\Rightarrow [e_i, \sum_{j=1}^n x_j e'_j] = 0$, $i = 1, \ldots, n$

Has no nontrivial solns.

$\Rightarrow \sum_{j=1}^n x_j (d'_j + 1) c_{ij}^k = 0$, $1 \leq i, k \leq n$

has no nontrivial solution

$\Rightarrow A_{d' + \delta_j} = 0 \Rightarrow A_d = 0$ for all $d \neq 0$ ∎